CERTIFY
TECHNOLOGY

Presented By: Certify Team

CERTIFY TECHNOLOGY

Innovating the Future with Al, App Development, and Full-Stack Solutions

Machine Learning Developer Internship level 1

- Experience: Gained foundational knowledge in machine learning, data preprocessing, and model development.

- Impact: Assisted in AI projects by implementing basic algorithms, analyzing datasets, and improving model performance.

- Skills Gained: Python, NumPy, Pandas, Scikit-Learn, Data Visualization, and Introductory Deep Learning Concepts.

Task 1 – Solve Any 2 Coding Challenge

Challenge 1: Implement a basic linear regression model using `sklearn` on a dataset (like predicting house prices).

Dataset :- You can Use your Own Dataset

Challenge 2: Use Pandas to clean and analyze a dataset (find missing values, duplicates, etc.).

DataSet:- https://www.kaggle.com/datasets/y asserh/titanic-dataset

Challenge 3: Implement a simple decision tree classifier on a small dataset (like Iris).

Dataset:- https://www.kaggle.com/datasets/pablomgomez21/drugs-a-b-c-x-y- for-decision-trees

Task 2: Build a Simple ML Project (Compulsory)

Project: Spam Email Classifier

- Train a machine learning model to classify emails as spam or not spam using the Naive Bayes algorithm.
- Steps:
    1) Preprocess the text data (convert emails into numerical format using TfidfVectorizer)
    2) Split data into training and testing sets.
    3) Train a Naive Bayes classifier and test its accuracy

- **DataSet:-** https://www.kaggle.com/datasets/balaka18/email-spam-classification-dataset-csv

Machine Learning Internship level 2

- Experience: Developed expertise in data collection, preprocessing, and exploratory data analysis using advanced statistical techniques.

- Impact: Contributed to data-driven decision-making by analyzing complex datasets, deriving meaningful insights, and designing impactful visualizations.

- Skills Gained: Proficiency in Python, Pandas, NumPy, SQL, and data visualization (Matplotlib, Seaborn). Gained hands-on experience in statistical modeling and feature engineering.

Task1:Machine Learning Prediction Project(Compulsory)

Challenge 1:- Insurance Charges Prediction Using Machine Learning

- Goal: Predict insurance charges based on client attributes such as age, BMI, smoking status, and other health factors.
- Techniques Used: Regression models (Linear Regression, Random Forest, XGBoost) for accurate prediction of insurance costs. Visualization techniques to understand the impact of each factor on charges.

  Dataset:- https://www.kaggle.com/datasets/thedevastator/prediction-of-insurance-charges-using-age-gender

Steps:-

- Load and preprocess the dataset
- Train different regression models (Linear Regression, Random Forest, XGBoost) Evaluate
- performance using MSE & R^2 score
- Deploy using Streamlit

Challenge 2:- Crime Rate Prediction Using SVM & Regression

- Goal: Predict crime rates based on historical data, location, and socio-economic factors.
- Techniques Used: SVM for classification +Regression models for crime rate forecasting
  Provide visualizations to show the impact of missing data.
- Dataset:- https://www.kaggle.com/datasets/sudhanvahg/indian-crimes-dataset

Steps:-

- Preprocess data (normalize crime categories, handle missing values).
- Train SVM model to classify crime-prone areas.
- Use Linear Regression or Random Forest to forecast crime trends
- Deploy using Streamlit (Easiest) or Flask API

Task 2: Machine Learning Project

Task 2's project is compulsory as it's the only one mentioned.

Challenge 1:- Retail Customer Segmentation using K-Means Clustering

- Goal: Create a K-Means clustering algorithm to group customers based on their purchase history and spending behavior.

- Techniques Used:
    - K-Means Clustering for customer segmentation
    - Elbow Method to determine the optimal number of clusters PCA or t-SNE for data visualization.
    - DataSet:- https://www.kaggle.com/datasets/yasserh/customer-segmentation-dataset

Steps:-

- Data Preprocessing
- Determine the Number of Clusters
- Apply K-Means Clustering Visualize the Clusters
- Deploy the Model

Machine Learning Internship level 3

- Experience: Gained expertise in deep learning, advanced model optimization, and scalable ML solutions. Worked on real-world datasets, fine-tuned models, and implemented AI-powered applications. Developed and deployed ML models using cloud-based platforms (AWS, GCP, or Azure).

- Impact: Contributed to data-driven decision-making by analyzing complex datasets, deriving meaningful insights, and designing impactful visualizations.

- Skills Gained: Proficiency in Python, Pandas, NumPy, SQL, and data visualization (Matplotlib, Seaborn). modeling and feature engineering.

Perform Any 1 Challenge

Challenge 1:- AI-Powered Fake News Detection System

Description: Build an ML model that can analyze news articles, social media posts, or blog content to detect fake news using NLP techniques.

Technology Used : Python, TensorFlow/PyTorch, NLP (BERT, TF-IDF), Flask/Django (for API), MongoDB/PostgreSQL

Key Features:

- Web scraping for real-time news analysis
- NLP-based text classification (Fake/Real)
- Explainable AI (why a news piece is fake)
- Deployment using FastAPI / Flask/Streamlit

DataSet: https://www.kaggle.com/datasets/emineyetm/fake-news-detection-datasets

Challenge 2:- Credit Card Fraud Detection System

Description: An ML model that detects fraudulent transactions in real time.

Technology Used: Python, Scikit-Learn, TensorFlow, Flask, MongoDB.

Key Features:

- Analyzes transaction data for fraud patterns.
- Uses anomaly detection techniques.
- Provides a fraud probability score for transactions.
- Can be integrated with banking systems for alerts.

Dataset:- https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

Challenge 3:- AI-Powered Personalized Movie Recommendation System

Description: Develop a system that recommends movies based on user preferences and watch history.

Technology Used: Python, Scikit-Learn, TensorFlow, Flask, MongoDB.

Key Features:

- Uses collaborative filtering for personalized recommendations.
- Analyzes user ratings and preferences.
- Provides a top 10 recommended movies list for each user.
- Integrates with IMDB/Netflix API for real-time movie data.

DataSet:- https://www.kaggle.com/datasets/parasharmanas/movie-recommendation-system

Submission Details:

- **File Format:** Submission Instructions (Google Form)
- Jupyter Notebook (.ipynb) or Python Script (.py)
- Contains the complete ML code, including data preprocessing, model training, and evaluation.
- Screenshot(s) of Output & Visualizations (.png or .jpg)
- Include key results such as graphs, accuracy scores, confusion matrices, etc.

Mastery in codinoft isn't just about writinoft lines of code; it's about architectinoft solutions, innovatinoft with purpose, and transforminoft ideas into impact. Elevate your journey—code with vision, build with precision.